

AP Project Diagrams

University ERP

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1 Use Case Overview

The ERP system supports three primary roles: Student, Instructor, and Admin. Tables 1–3 summarise the main use cases.

1.1 Student Use Cases

Use Case	Description
Login	Authenticate using username/password (auth_db) and load student profile (erp_db).
View Catalog	View list of Courses (Code, Title, Credits, Instructor, Seats).
Register	Register in a Section (checks: deadlines, time conflicts, course-unique rule, seat availability).
Drop Section	Drop an enrolled Section before the drop_deadline.
View Timetable	View weekly timetable built from registered Sections.
View Grades	View component scores (Quiz 20, Midsem 30, Endsem 50) and Final Grade.
Download Transcript	Export transcript as CSV/PDF.

Table 1: Student Use Cases

1.2 Instructor Use Cases

Use Case	Description
Login	Authenticate as Instructor and load instructor dashboard.
View My Sections	See only Sections assigned to the instructor.
Enter Scores	Input scores for Quiz (20), Midsem (30), Endsem (50). Validation: no negative or > max.
Class Stats	View Min/Max/Average per component.
Download Stats	Download .csv containing class performance.
Broadcast	Send announcement to all students enrolled in a section.

Table 2: Instructor Use Cases

1.3 Admin Use Cases

Use Case	Description
Login	Authenticate as Admin.
Manage Users	Create Students/Instructors/Admins; delete users.



Manage Courses	Create, edit, delete Course entries.
Manage Sections	Create Sections with deadlines, timing, capacity.
Assign Instructor	Attach an Instructor to a Section.
Maintenance Mode	Turn ON/OFF maintenance mode (blocks all writes).
Backup/Restore	Export or import SQL dumps for auth_db and erp_db.

Table 3: Admin Use Cases

2 Domain Model (“Things” Sketch)

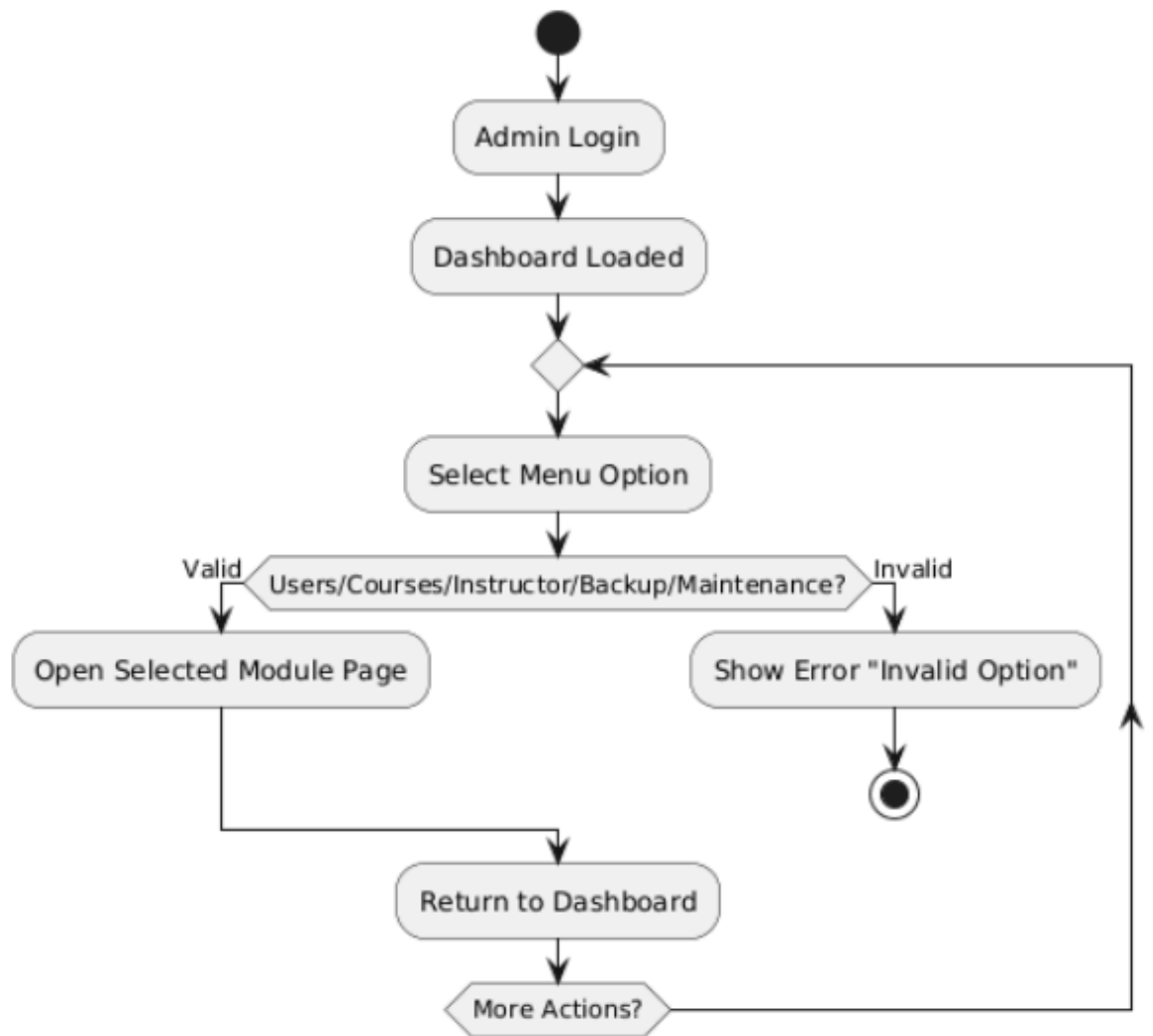


Figure 1: Admin Dashboard

3 Flow Diagrams

3.1 Register and Drop Courses

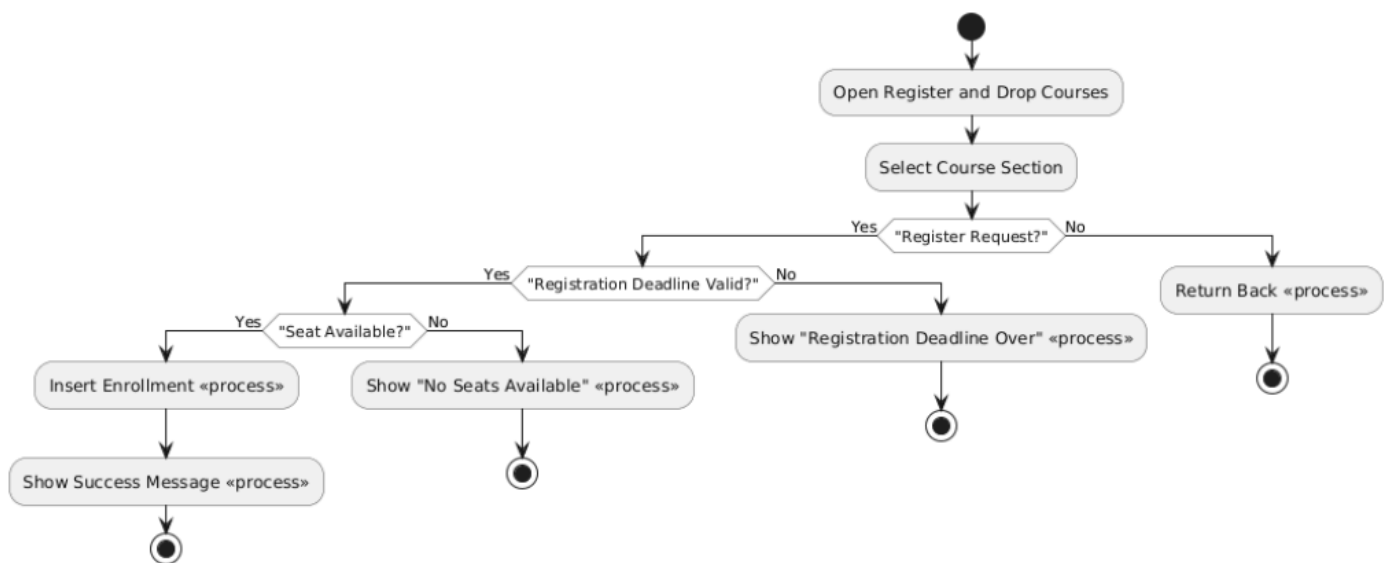


Figure 2: Activity flow for student course enrollment.

3.2 Grade Entry and Final Grade Computation Flow

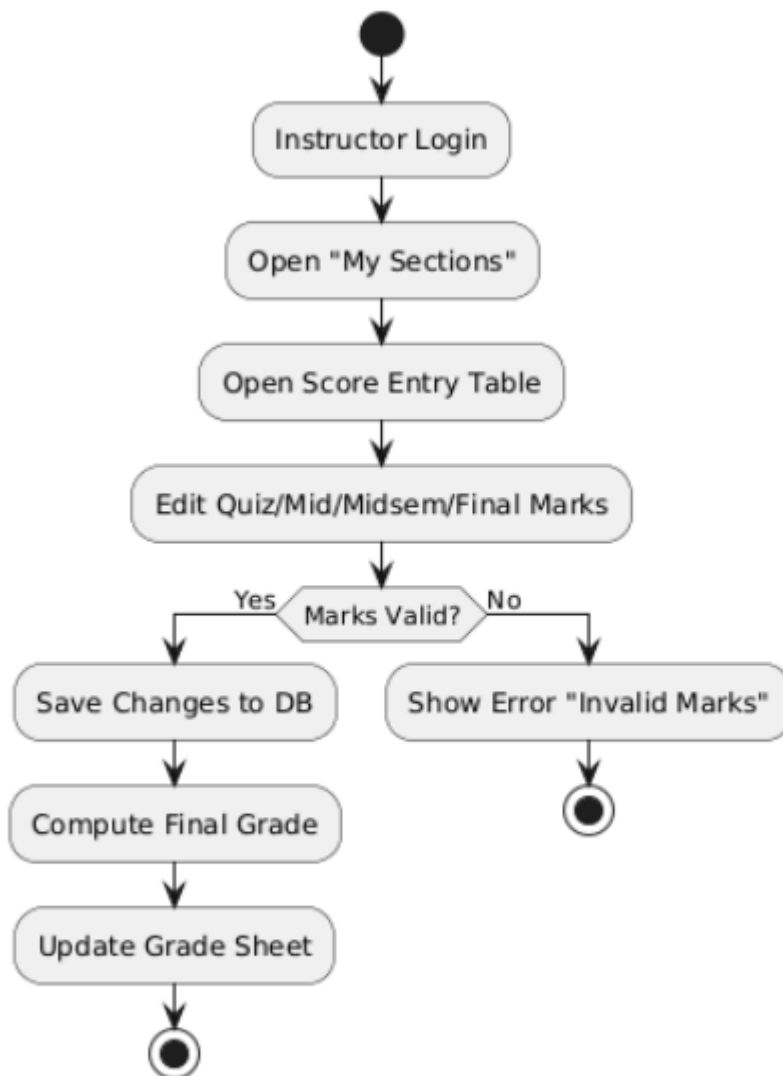


Figure 3: Activity flow for instructor grade entry and final grade.

3.3 Maintenance Mode Toggle Flow

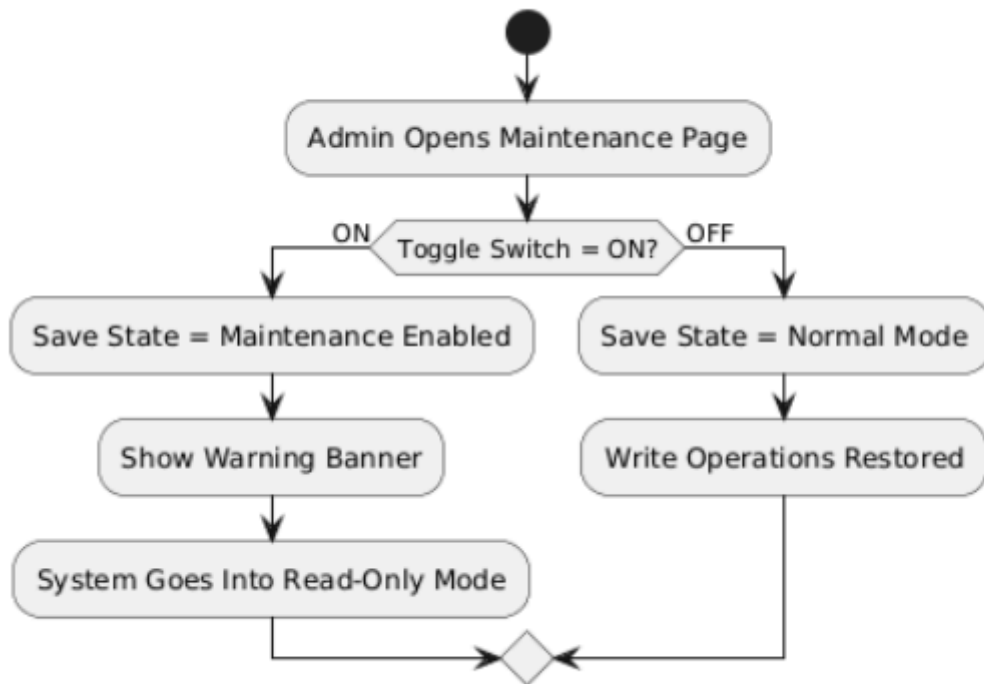


Figure 4: Activity flow for toggling system maintenance mode.